

AI-Powered Knowledge Base System

Project Overview

An AI-Powered Knowledge Base allows users to upload documents and ask questions in natural language. The system retrieves relevant information from company documents and uses an LLM to generate accurate answers.

Example:

User: "How many annual leaves can employees take?"

AI: "According to the HR policy document, employees are entitled to 24 annual paid leaves per year."

Enterprise Use Cases

HR Assistant

- Leave policies
- Company guidelines
- Employee handbook

IT Support Assistant

- Troubleshooting guides
- Internal documentation
- Infrastructure manuals

Legal Assistant

- Contracts
- Compliance documents
- Regulations

Customer Support Assistant

- Product documentation
 - FAQs
 - Support manuals
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System Architecture

Frontend (React/Angular)



Spring Boot REST API



Spring Security (JWT)



Document Service



Vector Database



Spring AI



LLM (OpenAI/Gemini/Ollama)

Core Features

Authentication

- Register
- Login
- JWT Security
- Role-based access

Roles:

- Admin
 - Employee
 - Viewer
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Document Management

Users can:

- Upload PDFs
- Upload DOCX files
- Upload TXT files
- Delete documents
- View uploaded documents

Metadata:

- File name
 - Upload date
 - Owner
 - Department
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Document Processing

When a document is uploaded:

1. Extract text
2. Split text into chunks
3. Generate embeddings
4. Store embeddings in vector database

Example:

PDF

↓

Text Extraction

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Chunking

↓

Embeddings

↓

Vector Storage

AI Chat

User asks:

"What is the refund policy?"

System:

1. Generate embedding of question
 2. Search vector database
 3. Retrieve relevant chunks
 4. Send context to LLM
 5. Generate answer
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Search

Support:

- Semantic Search
- Keyword Search
- Hybrid Search

Example:

Query: "Vacation policy"

Results:

- Leave Policy.pdf
 - Employee Handbook.pdf
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Chat History

Store:

- User messages
- AI responses
- Timestamps

Features:

- Previous conversations

- Search chats
 - Export history
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Citations

AI should provide sources.

Example:

Answer: "Employees are eligible for 24 annual leaves."

Sources:

- HR_Policy.pdf
 - Page 12
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Database Design

Users

- id
- name
- email
- password
- role

Documents

- id
- file_name
- owner_id
- upload_date

Embeddings

- id
- document_id
- chunk_text
- vector

Chats

- id
- user_id

- question
 - answer
 - timestamp
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Technology Stack

Backend:

- Java 21
- Spring Boot
- Spring Security
- Spring Data JPA

AI:

- Spring AI
- OpenAI / Gemini / Ollama

Database:

- PostgreSQL

Vector Database:

- pgvector
- Qdrant
- Weaviate

Infrastructure:

- Docker
- Docker Compose

Monitoring:

- Prometheus
 - Grafana
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Advanced Enterprise Features

Department Isolation

HR employees can only access HR documents.

Finance employees can only access Finance documents.

Audit Logs

Track:

- Uploads
 - Deletions
 - Searches
 - AI queries
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AI Summarization

Generate:

- Document summaries
 - Key points
 - Action items
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AI Document Comparison

Compare:

- Contract versions
 - Policy updates
 - Technical documents
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Multi-Document Chat

User:

"Compare leave policy and contractor policy."

AI searches multiple documents and produces a combined answer.

Resume Description

Developed an enterprise-grade AI Knowledge Base using Spring Boot, Spring Security, PostgreSQL, pgvector, and Spring AI. Implemented Retrieval-Augmented Generation (RAG), document ingestion,

semantic search, JWT authentication, role-based access control, source citations, and AI-powered document question answering.